

DSAA3073: Theories in Data Science

Week 4: Bias-Variance Tradeoff & Regularization

Concise Learning Sheet

Wei Wang · DSA, HKUST(GZ) · 2026-06-27

Opening Challenge: The Model Selection

This phenomenon—where complex models fit training data perfectly but fail on new data—was recognized as early as the 1960s but only formalized mathematically in the 1970s with the bias-variance decomposition.

🎯 Opening Challenge

The Housing Price Predictor

You're building a model to predict housing prices. You have three options:

1. **Simple:** $\text{price} = \beta_0 + \beta_1 \cdot \text{sqft}$
2. **Medium:** $\text{price} = \beta_0 + \beta_1 \cdot \text{sqft} + \beta_2 \cdot \text{bedrooms} + \beta_3 \cdot \text{location} + \dots$
3. **Complex:** 100 features including polynomials, interactions, neighborhood statistics

Observations:

- Model 1: Training error = 15%, Test error = 16%
- Model 2: Training error = 8%, Test error = 9%
- Model 3: Training error = 2%, Test error = 25%

Question: Which model should you deploy? Why does the most complex model perform worst on test data despite fitting training data best?

The answer lies in understanding the fundamental tradeoff between bias (systematic error) and variance (estimation instability). This week, we'll formalize this tradeoff and develop regularization as a principled solution.

✓ Checkpoint

Before we begin: In the opening challenge, Model 3 had 2% training error but 25% test error. What does this 23-percentage-point gap tell you about the model's behavior?

Answer: The large gap indicates the model is memorizing training data rather than learning generalizable patterns—a hallmark of overfitting.

The Bias-Variance Decomposition

Motivation: We've seen that complex models can fail spectacularly on test data despite perfect training performance. But why? The bias-variance decomposition provides a mathematical microscope to examine prediction error, revealing that it stems from three distinct sources: systematic model limitations (bias), sensitivity to random sampling (variance), and irreducible noise. Understanding this decomposition is the key to controlling model complexity.

← From Week 2: VC Theory

We learned that larger hypothesis classes (higher VC dimension) can fit more patterns but require more data to generalize. The bias-variance decomposition quantifies this tradeoff numerically: large classes have low bias but high variance.

Setup and Motivation

Consider the **regression problem** (the task of predicting a continuous output variable from input features): we observe (x_i, y_i) where $y_i = f^*(x_i) + \varepsilon_i$ with $\mathbb{E}(\varepsilon_i) = 0$ and $\text{Var}(\varepsilon_i) = \sigma^2$.

Here, $f^*(x)$ is the **true underlying function** (the ideal relationship between inputs and outputs without noise), ε_i is the **observation noise** (random error in measurements), and $\hat{f}_n(x)$ is our **estimator** (the function we learn from n training samples).

We train an estimator $\hat{f}_n(x)$ on n samples and want to predict at a new point x .

Example: Housing Prices - Setup

Throughout this section, imagine predicting house prices from square footage. We have $n = 50$ training houses, and the true relationship is roughly quadratic: $f^*(x) = 100 + 0.5x + 0.0001x^2$ (in thousands of dollars), but our observations have noise $\varepsilon_i \sim \mathcal{N}(0, 20^2)$.

Definition: Mean Squared Error

At a fixed input x , the **prediction error** (the expected squared difference between our prediction and the true value) is measured by the **Mean Squared Error** (MSE):

$$\text{MSE}(x) = \mathbb{E}\left(\left(y - \hat{f}_n(x)\right)^2\right) \quad (1)$$

where the expectation is over the training sample used to fit $\hat{f}_n$ and the fresh test noise in y . This expectation accounts for randomness from two sources: (1) which particular training samples we happened to observe, and (2) the noise in the new test observation y .

🤔 Pause and Think

Question: How would you decompose this error? What sources of error can you identify?

Hint: Add and subtract $f^*(x)$ and $\mathbb{E}(\hat{f}_n(x))$, then expand the square.

Answer: The key is to separate the error into: (1) bias from model assumptions, (2) variance from data sampling, and (3) irreducible noise.

The Fundamental Decomposition

Theorem: Bias-Variance Decomposition

At a fixed x , assume $y = f^*(x) + \varepsilon$ with $\mathbb{E}(\varepsilon | x) = 0$, $\text{Var}(\varepsilon | x) = \sigma^2$, and the fresh test noise ε independent of the training sample S . For any estimator $\hat{f}_n$ trained on S , the MSE (expectation over training samples S and fresh test noise ε) decomposes as:

$$\text{MSE}(x) = \underbrace{\left(\mathbb{E}_S(\hat{f}_n(x)) - f^*(x)\right)^2}_{\text{Bias}^2} + \underbrace{\text{Var}_S(\hat{f}_n(x))}_{\text{Variance}} + \underbrace{\sigma^2}_{\text{Irreducible}} \quad (2)$$

The three terms represent distinct error sources:

- **Bias squared** $\left(\mathbb{E}_S(\hat{f}_n(x)) - f^*(x)\right)^2$: systematic error from model limitations (e.g., fitting a line to curved data)
- **Variance** $\text{Var}_S(\hat{f}_n(x))$: prediction instability due to random training sample variation
- **Irreducible error** σ^2 : inherent noise in the data that no model can eliminate

Proof. Stage 1: Isolate irreducible error

Let $y = f^*(x) + \varepsilon$ where $\mathbb{E}(\varepsilon) = 0$. We decompose:

$$\text{MSE}(x) = \mathbb{E}\left((y - \hat{f}_n(x))^2\right) = \mathbb{E}\left((f^*(x) + \varepsilon - \hat{f}_n(x))^2\right) = \mathbb{E}\left((f^*(x) - \hat{f}_n(x))^2\right) + \mathbb{E}(\varepsilon^2) + 2\mathbb{E}(\varepsilon(f^*(x) - \hat{f}_n(x)))$$

The cross term vanishes since $\mathbb{E}(\varepsilon) = 0$ and ε is independent of $\hat{f}_n$. Thus:

$$\text{MSE}(x) = \mathbb{E}\left((f^*(x) - \hat{f}_n(x))^2\right) + \sigma^2 \quad (4)$$

Stage 2: Separate bias and variance

Now add and subtract $\mathbb{E}(\hat{f}_n(x))$:

$$= \mathbb{E}\left((f^*(x) - \mathbb{E}(\hat{f}_n(x)) + \mathbb{E}(\hat{f}_n(x)) - \hat{f}_n(x))^2\right) = \mathbb{E}\left((f^*(x) - \mathbb{E}(\hat{f}_n(x)))^2\right) + \mathbb{E}\left((\hat{f}_n(x) - \mathbb{E}(\hat{f}_n(x)))^2\right) + 2\mathbb{E}\left((f^*(x) - \mathbb{E}(\hat{f}_n(x))) (\hat{f}_n(x) - \mathbb{E}(\hat{f}_n(x)))\right)$$

The last term is zero since $\mathbb{E}(\hat{f}_n(x) - \mathbb{E}(\hat{f}_n(x))) = 0$. Therefore:

$$\text{MSE}(x) = \left[\mathbb{E}(\hat{f}_n(x)) - f^*(x)\right]^2 + \text{Var}[\hat{f}_n(x)] + \sigma^2 \quad (6)$$

■

🎯 Key Takeaway

The bias-variance decomposition reveals three sources of prediction error. Operationally, imagine repeatedly resampling the training set and refitting the model: bias measures systematic displacement from $f^*(x)$, while variance measures how much predictions move across refits.

1. **Bias:** Systematic error from model assumptions (e.g., fitting a line to nonlinear data)
2. **Variance:** Error from sensitivity to training data fluctuations
3. **Irreducible error:** Noise in the data that no model can eliminate

🧠 Mental Model: The Bias-Variance Seesaw

Simple Model	Optimal	Complex Model
High Bias	Sweet	High Variance
Low Variance	Spot	Low Bias
Underfits		Overfits

$$\text{Total Error} = \text{Bias}^2 + \text{Variance} + \sigma^2$$

Think of model selection as balancing on a seesaw: push toward complexity (right) and variance rises while bias falls; pull toward simplicity (left) and bias rises while variance falls. The optimal model is where total error is minimized—rarely at either extreme.

Example: Housing Prices - Bias vs Variance

- **Linear model** $\hat{f}(x) = \beta_0 + \beta_1 x$: Systematically underpredicts large houses (high bias), but predictions are stable across training sets (low variance).
- **Degree-15 polynomial**: Can fit any training set perfectly (low bias), but predictions swing wildly if we change one training house (high variance).

✓ Checkpoint

Question: If we have infinite training data ($n \rightarrow \infty$), which component of MSE vanishes?

Answer: Variance vanishes because with infinite data, $\hat{f}_n(x)$ converges to $\mathbb{E}(\hat{f}_n(x))$. Bias and irreducible error remain.

Concrete Example: Polynomial Regression

Example: Polynomial Fitting

Consider fitting polynomials of degree d to data from $f^*(x) = \sin(x)$.

The values below are illustrative of the typical qualitative pattern.

The “sweet spot” depends on sample size n , noise level, and the smoothness of f^* . With more data, we can often afford higher complexity, but there is no universal law such as one fixed scaling for the optimal degree.

Degree 1 (Linear):

- Bias: High (cannot capture sinusoidal shape)
- Variance: Low (only 2 parameters to estimate)
- $\text{MSE} \approx 0.25 + 0.02 + 0.09 = 0.36$

Degree 5:

- Bias: Low (flexible enough to approximate sine)
- Variance: Moderate (6 parameters)
- $\text{MSE} \approx 0.02 + 0.08 + 0.09 = 0.19 \leftarrow \text{Optimal}$

Degree 15:

- Bias: Very low (can fit almost anything)
- Variance: High (16 parameters, unstable)
- $\text{MSE} \approx 0.01 + 0.35 + 0.09 = 0.45$

← From Week 3: Rademacher Complexity

We proved that generalization error is bounded by $O\left(\sqrt{\frac{\log(|\mathcal{H}|)}{n}}\right)$ for finite hypothesis classes. Polynomial degree d directly controls $|\mathcal{H}|$ —higher degree means larger class, slower convergence, and higher variance for fixed n . The optimal degree balances approximation (bias) and estimation (variance).

🤔 Pause and Think

Question: If we double the sample size from $n = 50$ to $n = 100$, how might the optimal polynomial degree change?

Hint: Variance decreases with more data, so a somewhat richer model may become viable.

Answer: It may increase, but the amount depends on the unknown signal smoothness, noise level, sampling design, and criterion used. The qualitative lesson is the bias-variance tradeoff, not a universal formula for d^* .

📊 Confidence Check

Self-Assessment: Can you...

1. Derive the bias-variance decomposition from scratch, explaining each step?
2. Given a specific estimator and data distribution, compute bias and variance?
3. Explain why bias and variance typically move in opposite directions as model complexity changes?

If unsure on any, revisit the theorem and proof before continuing.

🔧 Construct Your Own

Design Your Own Example

Create a regression scenario where:

1. Choose a true function $f^*(x)$ (can be piecewise, trigonometric, polynomial)
2. Pick two model classes: one too simple (high bias), one too complex (high variance)
3. For each model, sketch qualitatively how bias and variance change with sample size n
4. Identify the “crossover” point where the complex model becomes preferable

Extension: How does noise level σ^2 affect this crossover?

Progress on backbone: The bias-variance decomposition reveals that prediction error splits into three components. We can’t eliminate irreducible noise, but we **can** control the bias-variance tradeoff. The question becomes: how do we systematically reduce variance without accepting unlimited bias?

We’ve seen that controlling variance is crucial, but **why** does variance become problematic in the first place? The answer lies in a deep mathematical insight: when we fit complex models to noisy data, we’re solving an **inverse problem**—and not all such problems are well-behaved. Understanding this connection transforms model selection from an art into a science.

Ill-Posed Inverse Problems

Jacques Hadamard (1902) introduced these criteria while studying partial differential equations. He distinguished “physical” problems (well-posed) from “mathematical” ones (possibly ill-posed).

Motivation: When we fit a complex model to noisy data, we’re s given noisy outputs, recover the underlying function. But not all problems are created equal. Hadamard identified three requirements for a problem to be “well-posed,” and model selection often violates the third—stability. Without stability, tiny data changes cause huge model changes, which is precisely the high-variance behavior we want to avoid. Regularization is the mathematical tool that restores stability.

Hadamard’s Well-Posedness Criteria

Definition: Well-Posed Problem (Hadamard)

A problem is **well-posed** (mathematically stable and physically meaningful) if it satisfies three criteria:

1. **Existence:** A solution exists for every admissible input
2. **Uniqueness:** The solution is unique (no ambiguity)
3. **Stability:** The solution depends continuously on the input data (small input changes $\rightarrow$ small output changes)

If any criterion fails, the problem is **ill-posed** (unstable or ambiguous). Model selection typically fails the stability criterion: tiny changes in noisy data can lead to drastically different model choices.

Why Model Selection is Ill-Posed

Consider the **inverse problem** (the task of recovering unknown parameters from observed data): given noisy observations π^δ , find model parameters a such that $\Pi(a) \approx \pi^\delta$.

Here, Π is the **forward operator** (the function that maps parameters to predictions), π^δ represents **noisy measurements**, and δ is the **noise level**.

✗ Wrong Path: Naive Approach

Minimize $\|\Pi(a) - \pi^\delta\|^2$ without constraints.

Problem: Multiple models fit equally well within noise level δ . Small data perturbations lead to drastically different selected models. This violates Hadamard’s stability criterion.

Example: Noisy Polynomial Interpolation

Problem: Recover a smooth curve from noisy measurements.

Setup:

- True signal: an unknown smooth function $f^* : [0, 1] \rightarrow \mathbb{R}$
- Observations: $y_i = f^*(x_i) + \varepsilon_i$ at grid points x_i
- Model: a degree-20 polynomial p chosen to fit the observations
- Goal: find a curve p with $p(x_i) \approx y_i$ and good predictions between sample points

Naive exact interpolation solves:

$$p^* = \arg \min_{p \in \mathcal{P}_{20}} \sum_{i=1}^n (p(x_i) - y_i)^2 \quad (7)$$

and may drive the training error nearly to zero.

Issues:

1. Non-existence/overconstraint: exact interpolation may be impossible if the model class is restricted
2. Non-uniqueness: many high-degree curves can fit noisy data almost equally well
3. Instability: small perturbations in one y_i can create large oscillations between sample points

Consequence: The fitted curve can change dramatically after adding tiny noise to the labels, even though the observed data look almost unchanged.

 Connection

Link to PAC Learning: Ill-posedness is related to overfitting. Without regularization, we're selecting from an overly complex hypothesis class, leading to high variance (instability).

← **From Week 3: Uniform Convergence**

We proved that empirical risk converges uniformly to true risk when the hypothesis class has finite complexity. But convergence is slow for large classes—leading to high variance for finite n . Regularization effectively shrinks the hypothesis class to accelerate convergence.

✓ **Checkpoint**

Question: In Hadamard's framework, which criterion does overfitting violate: existence, uniqueness, or stability?

Answer: Stability. Many models fit the training data equally well, but small noise perturbations select drastically different models.

Example: Housing Prices - Instability

Suppose we fit a degree-20 polynomial to our 50 houses. If house #17's price changes by \$1000 (within noise level), the fitted polynomial might develop wild oscillations in regions far from house #17. This is instability—the hallmark of ill-posedness.

Progress on backbone: Now we see the deep connection: high variance in the bias-variance decomposition corresponds to instability in Hadamard's framework. Both are symptoms of the same underlying disease—ill-posedness. But this diagnosis suggests a cure.

Hadamard didn't just identify the problem—mathematicians developed a principled solution. Tikhonov regularization transforms ill-posed problems into well-posed ones by adding a penalty for solution complexity. This is where the bias-variance tradeoff becomes **actionable**: we deliberately accept a small, controlled amount of bias to purchase a large reduction in variance (stability).

Tikhonov Regularization Framework

Andrey Tikhonov (1963) developed this framework for solving ill-posed problems in geophysics and inverse scattering. The same ideas were independently discovered in statistics as “ridge regression” (Hoerl & Kennard, 1970).

Motivation: We need a systematic way to trade bias for variance. To do this by adding a penalty term to the objective: instead of minimizing (which is ill-posed), we minimize data misfit plus a measure of solution complexity. The regularization parameter α controls the tradeoff—it’s the dial that moves us along the bias-variance seesaw toward stability.

General Formulation

Definition: Tikhonov Regularization

Let $\mathcal{H}$ be a **Hilbert space** (a complete vector space with an inner product, introduced in Week 1), $\Pi : \mathcal{A} \subseteq \mathcal{H} \rightarrow \mathbb{R}^d$ a (possibly nonlinear) **operator** (a function mapping between function spaces), and π^δ noisy observations with $\|\pi - \pi^\delta\| \leq \delta$.

A standard Tikhonov formal template is:

$$a_\alpha^\delta = \arg \min_{a \in \mathcal{A}} \left[\|\Pi(a) - \pi^\delta\|^2 + \alpha \|a - a_0\|_{\mathcal{H}}^2 \right] \quad (8)$$

where:

- $\alpha > 0$: **regularization parameter** (controls bias-variance tradeoff)
- $a_0 \in \mathcal{H}$: **prior** or **initial guess** (our belief about the solution before seeing data)

Existence of a minimizer requires additional closedness/compactness assumptions. In this sheet, the template mainly motivates finite-dimensional ridge regression below.

The objective has two competing terms:

1. **Data fidelity** $\|\Pi(a) - \pi^\delta\|^2$: measures how well the solution fits observed data (low value = good fit)
2. **Regularization** $\alpha \|a - a_0\|^2$: penalizes solutions far from the prior a_0 , enforcing simplicity/smoothness (low value = simple model)

Example: Housing Prices - Tikhonov Trust Region

Think of Tikhonov as a “trust region” approach: we trust the data (first term) but not completely—we also trust a prior belief a_0 (second term). The parameter α sets how much we trust the prior vs. the data.

- α small: “I trust my data”—fit closely, risk instability
- α large: “I trust my prior”—ignore data, high bias
- α optimal: balanced trust—best generalization

 Pause and Think

Question: What happens as $\alpha \rightarrow 0$? As $\alpha \rightarrow \infty$?

Hint: Consider the two terms in the objective function and which dominates in each limit.

Answer: $\alpha \rightarrow 0$: Unregularized solution (ill-posed). $\alpha \rightarrow \infty$: $a_\alpha^\delta \rightarrow a_0$ (ignore data completely).

Theoretical Guarantees

The next statements are qualitative inverse-problem facts. Their full versions require technical compactness, closedness, and source-condition assumptions; ridge regression below is the finite-dimensional case we will actually compute.

Theorem: Stability (Schematic)

Under standard compactness and continuity assumptions, for fixed $\alpha > 0$, Tikhonov minimizers depend continuously on the observed data.

Interpretation: Small data perturbations lead to small changes in the regularized solution.

Theorem: Convergence (Schematic)

If the noise level tends to zero and the regularization parameter α also tends to zero at a compatible rate, Tikhonov solutions approach an appropriate exact solution, often the solution closest to the prior a_0 .

Theorem: Convergence Rate (Schematic)

Under additional smoothness/source conditions, one can prove quantitative convergence rates. The precise rate depends on the source condition and the choice of α .

 Key Takeaway

Tikhonov regularization transforms an ill-posed problem into a well-posed one. The price is bias (solution differs from unregularized), but the gain is stability (continuous dependence on data).

Choosing the Regularization Parameter

Definition: Discrepancy Principle (Morozov)

Select the largest α such that:

$$\|\Pi(a_\alpha^\delta) - \pi^\delta\| \leq \tau\delta \quad (9)$$

where $\tau > 1$ is a safety factor (typically $\tau \in [1.1, 2]$).

Intuition: Don't fit data better than the noise level allows.

 Pause and Think

Question: Why choose the largest α satisfying the constraint, not the smallest?

Hint: Think about what larger α means for solution stability.

Answer: Larger α means more regularization $\rightarrow$ more stable solution. We want maximum stability subject to fitting data within noise level.

 $\rightarrow$ Week 5 Preview: Cross-Validation

The discrepancy principle requires knowing the noise level δ , which we rarely know in practice. Next week, we'll learn cross-validation—a data-driven method to choose α (or λ) without knowing δ or using test data.

 Construct Your Own**Explore Regularization Strength**

Consider the regularized objective $\|\Pi(a) - \pi^\delta\|^2 + \alpha\|a\|^2$.

1. For $\alpha \in \{0.01, 0.1, 1, 10, 100\}$, sketch qualitatively how the solution a_α changes
2. Identify which α values lead to: (a) overfitting, (b) underfitting, (c) good generalization
3. How would you visualize the “regularization path” (solution vs. α)?

Challenge: Can you prove that $\|a_\alpha\|$ is monotonically decreasing in α ?

Progress on backbone: The Tikhonov framework shows us **how** to regularize—add a penalty term scaled by α . But the framework is abstract, working in infinite-dimensional function spaces. To make this practical, we need to specialize to the most important case in applied statistics.

The Tikhonov framework is elegant but abstract. To make it concrete and computationally tractable, we now specialize to the workhorse of statistical learning: linear regression with quadratic penalties. This is **ridge regression**—simple enough to have closed-form solutions, yet powerful enough to be used everywhere from genomics to finance to machine learning.

Ridge Regression

Motivation: Ridge regression is Tikhonov regularization applied to linear least squares. It's the simplest nontrivial regularized method, yet it's used everywhere—from genomics to finance to machine learning. Why? Because it has a closed-form solution, clear geometric interpretation, and serves as a building block for more sophisticated methods (kernel ridge regression in Week 6, elastic net, etc.). Master ridge, and you've mastered the core ideas of regularization.

Problem Setup

Consider linear regression: $y = X\beta + \varepsilon$ where:

- $X \in \mathbb{R}^{n \times d}$: **design matrix** (n rows for samples, d columns for features)
- $\beta \in \mathbb{R}^d$: **parameters** or **coefficients** (weights for each feature)
- $\varepsilon \sim \mathcal{N}(0, \sigma^2 I)$: **noise** (random errors, normally distributed with variance σ^2)

Definition: Ordinary Least Squares (OLS)

Ordinary Least Squares (OLS) finds parameters that minimize squared prediction error on training data. If $X^\top X$ is invertible, the OLS estimator is:

$$\hat{\beta}_{\text{OLS}} = \arg \min_{\beta} \|y - X\beta\|^2 = (X^\top X)^{-1} X^\top y. \quad (10)$$

If $X^\top X$ is singular (not invertible), minimizers may not be unique and the **Moore-Penrose pseudoinverse** (a generalization of matrix inverse) gives the minimum-norm OLS solution.

Problems with OLS:

1. If $d > n$: $X^\top X$ is singular (no unique solution) - this is the **underdetermined** case where we have more features than samples
2. If $X^\top X$ nearly singular: $\|\hat{\beta}_{\text{OLS}}\|$ is huge (high variance) - small eigenvalues amplify noise
3. **Multicollinearity** (strong correlations between features): small data changes $\rightarrow$ large $\hat{\beta}$ changes

Example: Multicollinearity

Consider predicting GDP growth with correlated features:

- x_1 : inflation rate
- x_2 : unemployment rate
- x_3 : interest rate

If $x_1 \approx 2x_2 - x_3$ (nearly collinear), then $X^\top X$ has small eigenvalue. OLS produces:

$$\hat{\beta}_{\text{OLS}} = (100, -200, 100)^\top \quad (11)$$

(huge coefficients with opposite signs)

Small data perturbation changes this to:

$$\hat{\beta}_{\text{OLS}'} = (-50, 100, -50)^\top \quad (12)$$

(completely different!)

Example: Housing Prices - Correlated Features

Suppose we add correlated features: x_1 = square feet, x_2 = number of rooms (highly correlated with x_1), x_3 = lot size. OLS might produce:

- $\beta_1 = 500$ (per sqft), $\beta_2 = -30000$ (per room)

This doesn't make sense: adding a room **decreases** price? No—OLS is unstable. Ridge produces:

- $\beta_1 = 150$, $\beta_2 = 5000$ (sensible signs and magnitudes)

** Productive Struggle****Before reading the solution...**

Try deriving the ridge estimator yourself:

1. Write the objective $J(\beta) = \|y - X\beta\|^2 + \lambda\|\beta\|^2$
2. Expand in terms of transposes: $(y - X\beta)^\top (y - X\beta) + \lambda\beta^\top \beta$
3. Take gradient with respect to β and set to zero
4. Solve for β

Hint: You'll need $\nabla_\beta(\beta^\top A\beta) = 2A\beta$ for symmetric A .

Spend 5-10 minutes working through this before checking the proof below.

Ridge Regression Solution

Definition: Ridge Regression

Ridge regression adds an ℓ^2 penalty (sum of squared coefficients) to ordinary least squares:

$$\hat{\beta}_\lambda = \arg \min_{\beta} [\|y - X\beta\|^2 + \lambda \|\beta\|^2] \quad (13)$$

where $\lambda > 0$ is the **regularization parameter** (controls the strength of the penalty; larger λ = more regularization = smaller coefficients).

The two terms represent competing goals: fit the data well (first term) vs. keep coefficients small (second term).

Theorem: Closed-Form Solution

For $\lambda > 0$, the ridge estimator is:

$$\hat{\beta}_\lambda = (X^\top X + \lambda I)^{-1} X^\top y \quad (14)$$

Proof. The objective is:

$$J(\beta) = \|y - X\beta\|^2 + \lambda\|\beta\|^2 = (y - X\beta)^\top (y - X\beta) + \lambda\beta^\top \beta \quad (15)$$

Taking the gradient with respect to β (using the matrix calculus identities $\nabla_\beta(a^\top \beta) = a$ and $\nabla_\beta(\beta^\top A \beta) = 2A\beta$ for symmetric A):

$$\nabla_\beta J = -2X^\top y + 2X^\top X\beta + 2\lambda\beta = 0 \quad (16)$$

Solving for β by setting the gradient to zero (this is a necessary condition for a minimum):

$$(X^\top X + \lambda I)\beta = X^\top y \implies \beta = (X^\top X + \lambda I)^{-1} X^\top y \quad (17)$$

The **Hessian** (matrix of second derivatives) is $\nabla^2 J = 2(X^\top X + \lambda I) \succ 0$ (positive definite for $\lambda > 0$), confirming this is a global minimum. ■

 Key Takeaway

Ridge regression is always well-defined for $\lambda > 0$, even when $d > n$ or $X^\top X$ is singular. The matrix $X^\top X + \lambda I$ is guaranteed to be invertible. By contrast, OLS requires $X^\top X$ to be full rank (equivalently, X must have linearly independent columns).

 Checkpoint

Question: Why does adding λI to $X^\top X$ guarantee invertibility?

Answer: Because λI shifts all eigenvalues by λ . If $X^\top X$ has eigenvalues $\mu_1, \dots, \mu_d \geq 0$, then $X^\top X + \lambda I$ has eigenvalues $\mu_1 + \lambda, \dots, \mu_d + \lambda > 0$ (all positive, hence invertible).

→ Week 6 Preview: Kernel Ridge Regression

The formula $\hat{\beta}_\lambda = (X^\top X + \lambda I)^{-1} X^\top y$ extends beautifully to infinite-dimensional spaces. In Week 6, we'll see how the “kernel trick” allows ridge regression in reproducing kernel Hilbert spaces (RKHS) without ever computing in the high-dimensional space explicitly.

Progress on backbone: We have the ridge formula and know it's always well-defined. But **how** does ridge reduce variance? The answer lies in understanding its shrinkage behavior—ridge doesn't shrink all coefficients equally, it selectively shrinks based on the eigenstructure of the data.

Understanding ridge's properties reveals why it works. Ridge performs **selective shrinkage**: it shrinks coefficients corresponding to directions of low variance in the data (small eigenvalues of $X^\top X$) while preserving coefficients in high-variance directions (large eigenvalues). This is exactly what we want—noisy directions get dampened, stable directions preserved.

Properties of Ridge Estimator

Theorem: Shrinkage Property

When OLS exists, ridge shrinks each principal component of the

$$\hat{\beta}_\lambda = \sum_{j=1}^d \left(\frac{\mu_j}{\mu_j + \lambda} \right) u_j u_j^\top \hat{\beta}_{\text{OLS}}, \quad 0 \leq \frac{\mu_j}{\mu_j + \lambda} \leq 1$$

Hence $\|\hat{\beta}_\lambda\| \leq \|\hat{\beta}_{\text{OLS}}\|$. Strict shrinkage holds when $\hat{\beta}_{\text{OLS}}$ has a positive-eigenvalue direction.

The shrinkage factors $\frac{\mu_j}{\mu_j + \lambda}$ reveal how ridge works:

- Large eigenvalues μ_j : factor ≈ 1 (minimal shrinkage)
- Small eigenvalues μ_j : factor ≈ 0 (heavy shrinkage)

Ridge “fixes” ill-conditioning by shrinking directions with small eigenvalues. Small eigenvalues correspond to feature directions weakly supported by the data, so OLS must use large coefficients to fit them.

Proof. Let $X^\top X = U M U^\top$ (eigendecomposition, where $M = \text{diag}(\mu_1, \dots, \mu_d)$ contains the eigenvalues). Then:

$$\begin{aligned} \hat{\beta}_\lambda &= (U M U^\top + \lambda I)^{-1} X^\top y \\ &= U (M + \lambda I)^{-1} U^\top X^\top y \\ &= U (M + \lambda I)^{-1} M U^\top \hat{\beta}_{\text{OLS}} \\ &= \sum_{j=1}^d \left(\frac{\mu_j}{\mu_j + \lambda} \right) u_j u_j^\top \hat{\beta}_{\text{OLS}} \end{aligned} \tag{19}$$

Since $\frac{\mu_j}{\mu_j + \lambda} \leq 1$ for all j , we have $\|\hat{\beta}_\lambda\| \leq \|\hat{\beta}_{\text{OLS}}\|$, with strict inequality when a nonzero OLS component is actually shrunk. ■

🧠 Mental Model: Ridge as Selective Shrinkage

Imagine $X^\top X$ has eigenvalues $\mu_1 = 100, \mu_2 = 10, \mu_3 = 0.1$.

With $\lambda = 1$:

- Direction 1: shrinkage factor = $\frac{100}{100+1} \approx 0.99$ (almost no shrinkage)
- Direction 2: shrinkage factor = $\frac{10}{10+1} \approx 0.91$ (moderate)
- Direction 3: shrinkage factor = $\frac{0.1}{0.1+1} \approx 0.09$ (heavy shrinkage)

Key insight: Ridge doesn't shrink uniformly—it shrinks most aggressively in directions the data doesn't “care about” (small eigenvalues). This is exactly where OLS variance is highest!

🏗️ Scaffolded Derivation (5 steps)

Derive the Bias Formula

Given: $\hat{\beta}_\lambda = (X^\top X + \lambda I)^{-1} X^\top y$ and $y = X\beta^* + \varepsilon$.

Step 1: Substitute y into the ridge formula.

$$\hat{\beta}_\lambda = (X^\top X + \lambda I)^{-1} X^\top (X\beta^* + \varepsilon)$$

Step 2: Distribute the inverse-transpose product.

$$\hat{\beta}_\lambda = (X^\top X + \lambda I)^{-1} X^\top X\beta^* + (X^\top X + \lambda I)^{-1} X^\top \varepsilon$$

Step 3: Take expectation (using $\mathbb{E}(\varepsilon) = 0$).

$$\mathbb{E}(\hat{\beta}_\lambda) = (X^\top X + \lambda I)^{-1} X^\top X\beta^*$$

Step 4: Compute bias by subtracting β^* .

$$\text{Bias} = \mathbb{E}(\hat{\beta}_\lambda) - \beta^* = \left[(X^\top X + \lambda I)^{-1} X^\top X - I \right] \beta^* \quad (23)$$

Step 5: Simplify the bracket (combine over common denominator).

$$\left[(X^\top X + \lambda I)^{-1} X^\top X - I \right] = [(X^\top X) - (X^\top X + \lambda I)](X^\top X + \lambda I)^{-1} \quad (24)$$

$$= -\lambda(X^\top X + \lambda I)^{-1} \quad (25)$$

Therefore: $\text{Bias} = -\lambda(X^\top X + \lambda I)^{-1} \beta^*$

What does the bias formula tell us?

$$\text{Bias} = -\lambda(X^\top X + \lambda I)^{-1} \beta^* \quad (26)$$

- **Proportional to λ :** More regularization $\rightarrow$ larger bias (systematically shrinks toward zero)
- **Negative direction:** Ridge underestimates all coefficients in the direction determined by $(X^\top X + \lambda I)^{-1} \beta^*$
- **Matrix factor:** $(X^\top X + \lambda I)^{-1}$ determines which coefficients get shrunk most (those corresponding to small eigenvalues of $X^\top X$)
- **Depends on β^* :** Bias vanishes if $\beta^* = 0$ (nothing to shrink)

Progress on backbone: We've seen qualitatively that ridge shrinks coefficients, reducing variance. But to truly understand the bias-variance tradeoff, we need quantitative formulas. How much bias does regularization introduce? How much variance does it eliminate? And most importantly, when does the tradeoff favor ridge over OLS?

Now we make the bias-variance tradeoff precise for ridge regression. We'll derive exact formulas showing that bias grows with λ (more regularization = more bias) while variance shrinks with λ (more regularization = less variance). The optimal λ balances these opposing forces, achieving lower MSE than OLS when the signal is smooth or the noise is high.

Bias-Variance Analysis

Fixed-X assumption

This theorem conditions on X being fixed (non-random). Expectations are only over the noise ε , not over different possible training samples.

This is the standard assumption in classical regression theory, though in learning theory we often treat X as random (sampled from a distribution). The fixed- X analysis still provides insights about the bias-variance tradeoff.

Theorem: Bias and Variance of Ridge

Conditional on fixed design matrix X (expectations over noise ε only), ε with $\mathbb{E}(\varepsilon) = 0$ and $\text{Cov}(\varepsilon) = \sigma^2 I$:

Bias:

$$\mathbb{E}(\hat{\beta}_\lambda) - \beta^* = -\left(\lambda(X^\top X + \lambda I)^{-1}\right)\beta^* \quad (27)$$

Variance:

$$\text{Cov}(\hat{\beta}_\lambda) = \sigma^2(X^\top X + \lambda I)^{-1}X^\top X(X^\top X + \lambda I)^{-1} \quad (28)$$

🤔 Pause and Think

Question: How does bias and variance change as λ increases?

Hint: Consider how the ridge solution moves away from OLS as λ grows.

Answer: Bias increases (solution moves away from β^*), variance decreases (solution becomes more stable). Total MSE is U-shaped in λ .

Theorem: Variance Reduction

For any $\lambda > 0$ and positive eigenvalues μ_j :

$$\text{tr}(\text{Cov}(\hat{\beta}_\lambda)) = \sigma^2 \sum_{j=1}^d \frac{\mu_j}{(\mu_j + \lambda)^2} < \sigma^2 \sum_{j=1}^d \frac{1}{\mu_j} = \text{tr}(\text{Cov}(\hat{\beta}_{\text{OLS}})). \quad (29)$$

Proof. Using eigendecomposition $X^\top X = U M U^\top$ where $M = \text{diag}(\mu_1, \dots, \mu_d)$:

$$\begin{aligned} \text{tr}(\text{Cov}(\hat{\beta}_\lambda)) &= \sigma^2 \text{tr}[(M + \lambda I)^{-1} M (M + \lambda I)^{-1}] \\ &= \sigma^2 \sum_{j=1}^d \frac{\mu_j}{(\mu_j + \lambda)^2} \\ &< \sigma^2 \sum_{j=1}^d \frac{1}{\mu_j} = \text{tr}(\text{Cov}(\hat{\beta}_{\text{OLS}})) \end{aligned} \quad (30)$$

The inequality holds since $\frac{\mu_j}{(\mu_j + \lambda)^2} < \frac{1}{\mu_j}$ for all $\lambda > 0$. To see why:

$$\begin{aligned} \frac{\mu_j}{(\mu_j + \lambda)^2} < \frac{1}{\mu_j} &\iff \mu_j^2 < (\mu_j + \lambda)^2 \\ &\iff \mu_j^2 < \mu_j^2 + 2\lambda\mu_j + \lambda^2 \\ &\iff 0 < 2\lambda\mu_j + \lambda^2 \end{aligned} \quad (31)$$

which holds for all $\lambda, \mu_j > 0$.

Confidence Check

Self-Assessment: Bias-Variance Tradeoff in Ridge

Can you:

1. Explain why ridge bias increases with λ (with formula)?
2. Explain why ridge variance decreases with λ (with formula)?
3. Sketch the U-shaped curve of MSE vs. λ ?
4. Identify which λ values lead to underfitting vs. overfitting?

These are core concepts—review the theorems if you're uncertain.

Ivanov-Tikhonov Equivalence

Tikhonov (penalized):

$$\min_{\beta} \left(\frac{1}{2} \|y - X\beta\|^2 + \lambda \|\beta\|^2 \right) \quad (32)$$

Ivanov (constrained):

$$\min_{\|\beta\|^2 \leq t} \frac{1}{2} \|y - X\beta\|^2 \quad (33)$$

These are equivalent via KKT conditions: for each $\lambda > 0$, there exists $t > 0$ giving the same solution. Ivanov connects ridge to structural risk minimization (explicitly controlling hypothesis complexity).

✓ Checkpoint

Question: If $\beta^* = 0$ (no true signal), what is the bias of ridge regression?

Answer: Zero! If $\beta^* = 0$, then $\text{Bias} = -\lambda(X^\top X + \lambda I)^{-1} \cdot 0 = 0$. Ridge is unbiased at the origin, which aligns with the prior $\beta \sim \mathcal{N}(0, \frac{\sigma^2}{\lambda} I)$ centered at zero.

Example: Housing Prices - Ridge Effect

Suppose the true model is $\text{price} = \beta_0 + \beta_1 \cdot \text{sqft}$ (2 parameters), but we include 20 features (many irrelevant).

- **OLS:** Variance dominated by the 18 irrelevant directions—huge instability.
- **Ridge ($\lambda = 10$):** Shrinks irrelevant coefficients near zero, variance drops dramatically, small bias on the 2 true coefficients.
- **Ridge ($\lambda = 1000$):** All coefficients shrunk near zero, low variance but high bias even on true coefficients—underfitting.

Progress on backbone: We've now seen ridge regression from the bias-variance perspective—it reduces variance by shrinking coefficients, especially in directions with small eigenvalues. But there's a deeper way to understand what ridge is doing.

We've formulated ridge as a penalized optimization problem (Tikhonov formulation), but there's an equivalent constrained formulation (Ivanov formulation). Understanding **both** perspectives deepens intuition: Tikhonov says “minimize error plus penalty,” while Ivanov says “minimize error subject to a complexity budget.” These are two sides of the same coin, connected by Lagrange duality.

Practical Guidelines and Application

Ridge regression is particularly effective in genomics because biological systems exhibit “distributed effects”: many genes contribute small amounts rather than a few genes having large effects.

Motivation: Theory tells us that ridge works, but implementation is not scale-invariant, so standardization is essential. The optimal λ depends on unknown quantities, so we need data-driven selection methods (Week 5). This section bridges theory and practice, preparing you to use ridge regression correctly on real data.

Implementation Checklist

1. **Standardize features:** Ridge is not scale-invariant. Standardize X to have mean 0 and variance 1.
2. **Regularization path:** Compute $\hat{\beta}_\lambda$ for a grid of λ values (e.g., $\lambda \in [10^{-4}, 10^4]$ on log scale).
3. **Monitor shrinkage:** Plot $\|\hat{\beta}_\lambda\|$ vs. $\log(\lambda)$ to visualize coefficient decay.
4. **Cross-validation:** Use CV to select optimal λ (Week 5).
5. **Degrees of freedom:** Track effective complexity $\text{df}(\lambda) = \text{tr}[2(X^\top X + \lambda I)^{-1} X^\top X]$.

The degrees of freedom $\text{df}(\lambda)$ measures the effective number of parameters. As $\lambda \rightarrow 0$, $\text{df}(\lambda) \rightarrow d$ (OLS uses all d parameters). As $\lambda \rightarrow \infty$, $\text{df}(\lambda) \rightarrow 0$ (ridge shrinks to zero). Always $\text{df}(\lambda) \in [0, d]$.

Application: Gene Expression Prediction

Example: High-Dimensional Genomics

Problem: Predict disease from gene expression ($d = 10,000$ genes, $n = 100$ patients).

Challenge: $d \gg n$, so OLS is impossible.

Ridge solution:

$$\hat{\beta}_\lambda = (X^\top X + \lambda I)^{-1} X^\top y \quad (34)$$

Results (using a validation set for illustration; Week 5 explains how to choose λ without using test data):

- $\lambda = 0.1$: Test error = 0.23, $\|\hat{\beta}\| = 45$
- $\lambda = 1$: Test error = 0.18, $\|\hat{\beta}\| = 12 \leftarrow$ Optimal
- $\lambda = 10$: Test error = 0.21, $\|\hat{\beta}\| = 3$

Biological interpretation: Most genes have small effects (shrinkage is appropriate). Ridge provides stable predictions by pooling information across all genes.

✓ Checkpoint

Question: Why must we standardize features before applying ridge regression?

Answer: Because ridge penalizes $\|\beta\|^2 = \sum_j \beta_j^2$ uniformly. If feature j is measured in kilometers and feature k in millimeters, their natural coefficient scales differ by 10^6 —ridge would heavily shrink the millimeter coefficient even if both features are equally important. Standardization puts all features on equal footing.

Example: Housing Prices - Implementation

Implementation checklist:

1. Standardize: center and scale all features to mean 0, variance 1
2. Grid search: try $\lambda \in \{10^{-4}, 10^{-3}, \dots, 10^3, 10^4\}$
3. For each λ , compute 5-fold CV error (Week 5 explains how)
4. Select $\lambda^* = \arg \min_{\lambda} \text{CV error}$
5. Refit on all data with λ^* : $\hat{\beta}_{\lambda^*} = (X^\top X + \lambda^* I)^{-1} X^\top y$
6. Report test performance and coefficient magnitudes

Result: $\lambda^* = 10$, test MSE = 0.18, most stable coefficients.

🔧 Construct Your Own

Design a Ridge Regression Study

Choose a high-dimensional regression problem (or simulate one):

1. Generate data: $y = X\beta^* + \varepsilon$ with $n = 100$, $d = 50$, sparse β^*
2. Implement ridge: compute $\hat{\beta}_{\lambda}$ for $\lambda \in [10^{-3}, 10^3]$
3. Plot regularization path: $\beta_j(\lambda)$ for each $j = 1, \dots, d$
4. Track degrees of freedom: $\text{df}(\lambda) = \text{tr}[X(X^\top X + \lambda I)^{-1} X^\top]$
5. Compare ridge to OLS: MSE, coefficient stability, interpretability

Extension: How does performance change with $\frac{n}{d}$ ratio?

 Synthesis Challenge**Connecting Bias-Variance to VC Theory**

In Week 2, we learned that test error is bounded by training error plus a complexity term depending on $\text{VCdim}(\mathcal{H})$.

Task: Reconcile the VC bound with bias-variance decomposition:

1. How does $\text{VCdim}(\mathcal{H})$ relate to variance?
2. How does the approximation error (best-in-class error) relate to bias?
3. Show that the optimal hypothesis class balances these terms, just like optimal λ balances bias and variance.

Hint: The VC complexity term captures how much training error underestimates test error—this is exactly the variance component!

Transition: We've mastered the core theory and practice of ridge regression. The summary synthesizes everything and points forward to how ridge extends to more sophisticated settings.

Summary and Looking Ahead

Key Takeaway

Main Insights from Week 4:

1. **Bias-variance tradeoff:** Model complexity creates tension between systematic error (bias) and estimation instability (variance).
2. **Regularization as stabilization:** Tikhonov regularization transforms ill-posed inverse problems into well-posed ones by adding a penalty term.
3. **Ridge regression:** Closed-form solution $(X^\top X + \lambda I)^{-1} X^\top y$ shrinks coefficients, especially in directions with small eigenvalues.
4. **Equivalence:** Ivanov (constrained) and Tikhonov (penalized) formulations are equivalent via Lagrange duality.

Next week: How do we choose λ in practice? We will study cross-validation, information criteria (AIC/BIC), and other model selection techniques.

Connection to future topics:

- Week 6: Lasso (L1 penalty) for sparse solutions
- Week 8: Ridge in infinite-dimensional RKHS
- Week 11: Implicit regularization in deep learning

→ Week 5 Preview: Model Selection via Cross-Validation

We've derived ridge and understood its bias-variance tradeoff, but how do we choose λ without a validation set? Next week's cross-validation provides an elegant answer: partition the training data into folds, use each fold as a "mini test set" for the others, and select the λ that minimizes average fold error. This is the gold standard for tuning regularization parameters.

→ Week 6 Preview: Kernel Ridge Regression

Ridge regression extends beautifully to infinite dimensions via the kernel trick. The dual formulation $\hat{\alpha} = (K + \lambda I)^{-1} y$ (where K is the kernel matrix) allows ridge in RKHS without ever computing the feature map explicitly. This is how ridge powers support vector machines and Gaussian process regression.

Synthesis Challenge

Grand Challenge: Unifying Regularization and Generalization

Synthesize Weeks 2-4 by answering:

1. How does VC dimension relate to the bias-variance decomposition?
2. How does regularization effectively reduce VC dimension?
3. Why does the PAC bound improve with regularization?
4. Can you express the optimal λ in terms of n , d , σ^2 , and properties of β^* ?

Hint: Think of regularization as restricting the hypothesis class from $\mathcal{H}$ to $\mathcal{H}_\lambda = \{h : \|h\| \leq R(\lambda)\}$, where $R(\lambda)$ decreases with λ .

Mental Model: The Week 4 Conceptual Map

Bias-Variance Decomposition

↓

Complex models → High variance (instability)

↓

Ill-Posed Inverse Problems

↓

Tikhonov Regularization (add penalty)

↓

Ridge Regression (quadratic penalty)

↓

Shrinkage toward zero

↓

Reduced variance, increased bias

↓

Optimal λ balances bias-variance tradeoff

Every concept builds on the last: we diagnose the problem (bias-variance), identify its mathematical nature (ill-posedness), apply a general solution framework (Tikhonov), specialize to the most important case (ridge), and understand its effect (shrinkage). This logical flow is the backbone of Week 4.

Student Challenges

Student Challenge

1. Prove that ridge regression with $\lambda \rightarrow \infty$ converges to $\hat{\beta} = 0$.
2. Implement ridge regression and visualize the regularization path for a dataset with multicollinearity.
3. Derive the bias-variance decomposition for classification with 0-1 loss (harder than regression case).
4. Show that ridge is the MAP estimator under Bayesian prior $\beta \sim \mathcal{N}(0, \frac{\sigma^2}{\lambda} I)$.

▣ Proof Template: Proving Optimality of Ridge via Second Derivative Test

Goal: Show that $\hat{\beta}_\lambda = (X^\top X + \lambda I)^{-1} X^\top y$ minimizes $J(\beta) = \|y - X\beta\|^2 + \lambda \|\beta\|^2$.

Step 1: Compute first derivative $\nabla_\beta J$ and verify it vanishes at $\hat{\beta}_\lambda$.

Step 2: Compute second derivative (Hessian) $\nabla^2 J = \dots$ (fill in).

Step 3: Show the Hessian is positive definite for $\lambda > 0$ by checking eigenvalues.

Step 4: Conclude that $\hat{\beta}_\lambda$ is a global minimum (convexity ensures local = global).

Work through each step with explicit matrix computations.

▣ Proof Template: Deriving Variance of Ridge Estimator

Given: $\hat{\beta}_\lambda = (X^\top X + \lambda I)^{-1} X^\top y$ and $y = X\beta^* + \varepsilon$ with $\text{Cov}(\varepsilon) = \sigma^2 I$.

Goal: Show $\text{Cov}(\hat{\beta}_\lambda) = \sigma^2 (X^\top X + \lambda I)^{-1} X^\top X (X^\top X + \lambda I)^{-1}$.

Step 1: Write $\hat{\beta}_\lambda = (X^\top X + \lambda I)^{-1} X^\top (X\beta^* + \varepsilon)$.

Step 2: Note that variance comes only from the ε term (why?).

Step 3: Use $\text{Cov}(A\varepsilon) = A \text{Cov}(\varepsilon) A^\top$ with $A = (X^\top X + \lambda I)^{-1} X^\top$.

Step 4: Simplify: $A \text{Cov}(\varepsilon) A^\top = A(\sigma^2 I) A^\top = \sigma^2 A A^\top = \dots$

Complete the calculation.

 Theorem Landscape**Week 4 Theorem Hierarchy****Foundation:**

- Bias-Variance Decomposition (Theorem 1): Every estimator's MSE splits into $\text{bias}^2 + \text{variance} + \text{irreducible error}$

Application to Ridge:

- Ridge Closed-Form Solution (Theorem 2): $\hat{\beta}_\lambda = (X^\top X + \lambda I)^{-1} X^\top y$
- Shrinkage Property (Theorem 3): Ridge shrinks each principal component by factor $\frac{\mu_j}{\mu_j + \lambda}$
- Bias and Variance of Ridge (Theorem 4): Explicit formulas for bias and variance in terms of λ
- Variance Reduction (Theorem 5): Ridge always reduces variance compared to OLS (for $\lambda > 0$)

Equivalence:

- Ivanov-Tikhonov Duality (Theorem 6): Constrained and penalized formulations are equivalent via KKT

These six theorems form a complete story: decompose error $\rightarrow$ apply regularization $\rightarrow$ analyze bias-variance $\rightarrow$ prove equivalence of formulations.